

Smart Attendance

System Using
Facial Recognition

An Android/web-oriented approach to fast, accurate and accountable attendance management.

Live Camera → **Face Match** → **Attendance Record**

Review 2 • Complete Project Review

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FACE

RECOGNITION

ATTENDANCE

DATABASE

02 Objectives

What the solution is designed to achieve

Automate attendance

Replace repetitive manual marking with face-based verification.

Recognize in real time

Detect and identify registered users through a live camera.

Record accurately

Store the verified person, date and time automatically.

Reduce errors

Minimize missed entries, duplicate work and proxy attendance.

Secure & synchronize

Maintain user and attendance records in a centralized database.

Enable administration

Support user management, monitoring, reports and analytics.

DESIGN GOAL

Efficient • Accurate • Scalable

03 Problem Statement

Manual attendance creates operational friction and weakens record reliability

THE CURRENT REALITY

A simple daily task becomes a recurring administrative burden.

The challenge grows with every additional class, employee, register and reporting period.

Time

Calling names and entering data delays sessions.

Accuracy

Human error produces missing or inconsistent records.

Scale

Large groups make verification and follow-up difficult.

Control

Paper and spreadsheets limit monitoring and auditability.

NEED

A reliable digital process that identifies people and records attendance automatically.

04 Existing System

Typical manual workflow and its practical limitations

1. Call / sign

Teacher calls names or circulates a register.

2. Transcribe

Records move to paper sheets or spreadsheets.

3. Verify

Missing entries and disputes are checked manually.

4. Report

Totals and percentages are prepared later.

SLOW

Repeated every session

ERROR-PRONE

Missed or incorrect entries

PROXY RISK

Identity is not reliably verified

FRAGMENTED

Paper and files are hard to track

DELAYED

No live attendance view

MANUAL

Reports require extra effort

CENTRAL WEAKNESS

Manual identity confirmation.

05 Proposed System

Facial verification turns attendance into a controlled digital event

01 • REGISTER

Create a profile and capture facial data in advance.

02 • RECOGNIZE

Match the live face with a registered identity.

03 • RECORD

Write the attendance entry with date and time.

04 • REVIEW

Monitor records and generate reports from the dashboard.

EXPECTED OPERATIONAL CHANGE

Manual marking → Automatic verification | Delayed entry → Real-time recording

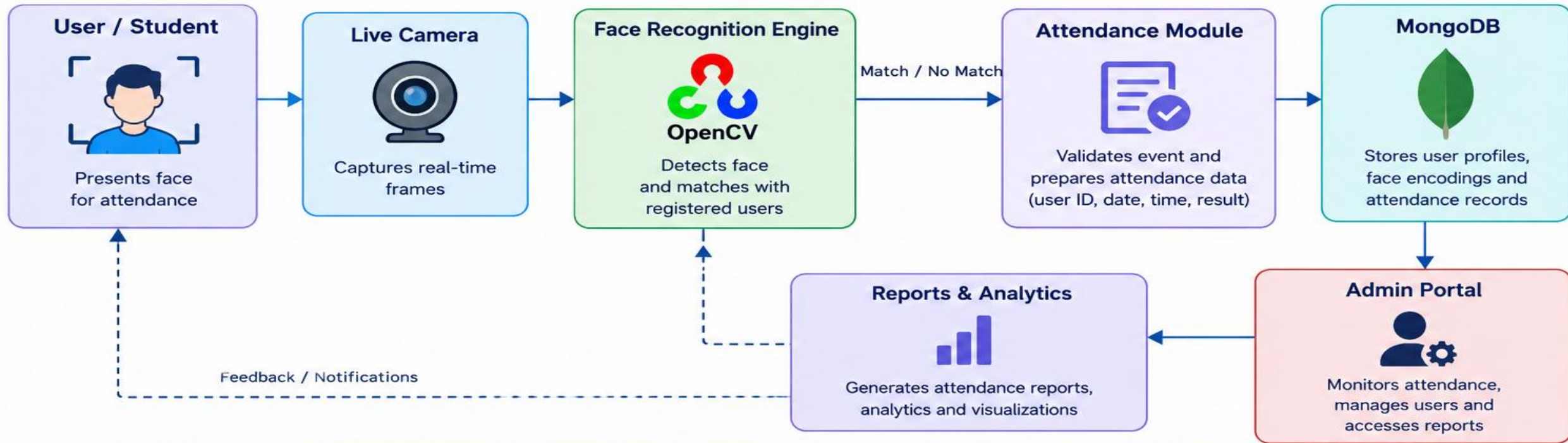
Paper / sheets → Central database | Manual totals → Automated reports

OUTCOME

Faster • More accurate • Easier to audit

System Architecture

Smart Attendance System Using Facial Recognition



Data Flow Summary



Technology Stack



07 Technology Stack

Each technology supports a distinct layer of the attendance workflow

INTERFACE

HTML • CSS • JavaScript — User-facing attendance and administration screens.

BACKEND

Python • Flask — Application logic, requests and service coordination.

FACE AI

OpenCV • Face Recognition — Detection, encoding comparison and identification.

DATABASE

MongoDB — Student profiles, face identifiers and attendance records.

CONNECTIVITY

PyMongo — Database access and record operations from Flask.

HARDWARE / DEV

Webcam or CCTV • Visual Studio Code — Real-time capture, coding, testing and development.

08 Database Design

MongoDB stores user profiles and attendance records centrally

USERS / STUDENTS

_id
name • roll_no / employee_id
department • face_encoding
created_at

ATTENDANCE

_id
user_id • date • time
status / verification result
optional image/reference path

ADMIN / MANAGEMENT

Administrative identity and access information can be maintained separately from attendance records.

ATTENDANCE EVENT

A verified match creates a timestamped event linked to the registered user.

RELATIONSHIP

One registered user → many attendance records

09 System Workflow

From registration to attendance record and administrative review

1. Registration

Capture/store face data

2. Live Capture

Get frame from camera

3. Face Detection

Locate face in frame

4. Recognition

Compare with registered identity

5. Verification

Confirm match

6. Record

Save timestamped entry

IF MATCHED

Create one timestamped attendance entry and show confirmation. Duplicate checks can prevent repeated marking in the same session.

IF UNCERTAIN

Do not mark automatically; request another scan or administrator review.

10 Key Features & Project Interface

Core capabilities mapped to everyday attendance operations

✓ **Real-time face recognition**

✓ **Automatic attendance marking**

✓ **User registration & management**

✓ **Admin dashboard with live view**

✓ **Reports and attendance analytics**

✓ **Centralized database records**

PROJECT OUTPUT

The system is designed to reduce manual marking effort, minimize errors, provide timestamped records and improve administrative visibility.

IMPORTANT

Use actual implementation screenshots during the final review to demonstrate the working system.

UI / DEMO EVIDENCE

Place your actual project screenshots here:

- Login / registration
- Camera / recognition screen
- Attendance confirmation
- Admin dashboard
- Attendance report

11 Testing, Results & Observations

Review-2 evidence should be demonstrated using the actual implemented system

TEST CASES

1. Registered face → verify identity
2. Unregistered face → reject / request review
3. Successful match → create attendance record
4. Repeat scan → prevent duplicate marking where implemented
5. Admin → view attendance records
6. Report → review attendance data

OBSERVABLE OUTPUT

- Face detected
- Identity matched / not matched
- Attendance timestamp created
- Record stored in MongoDB
- Admin can review records
- Reports can summarize attendance

EVIDENCE TO SHOW

Insert actual screenshots or live-demo captures:

- ✓ Recognition screen
- ✓ MongoDB record
- ✓ Dashboard
- ✓ Attendance report
- ✓ Test-case output

Do not add numerical accuracy claims unless measured in your project.

12 Conclusion, Limitations & Future Scope

Final project-review summary

CONCLUSION

The proposed system automates attendance through facial verification, timestamped records and centralized storage.

It addresses major weaknesses of manual attendance by reducing repetitive work and improving record visibility.

LIMITATIONS

- Recognition depends on camera/image quality
- Lighting and face angle can affect detection
- Uncertain matches require review
- Security and privacy controls need careful implementation
- Real-world accuracy should be validated with project test data

FUTURE SCOPE

- Mobile application support
- Liveness / anti-spoofing detection
- Integration with biometric devices
- AI-based attendance analytics & alerts
- Larger-scale deployment and testing

Thank You • Questions & Discussion